Chunyang Luo

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Research Interests: Agent Systems | Human-AI Interaction | LLM-based Planning & Reasoning | Multi-Robot Planning

EDUCATION

B.Eng. Candidate in Robotics Engineering, Peking University 2023 – Present
Expected Graduation: June 2027; GPA: 3.526/4.0 (85/100); Rank: 5/19 (Top 26%)
Selected Core Courses: Foundations of Machine Learning (98), Probability and Statistics (94), Introduction to Robotics (89)

PUBLICATIONS & MANUSCRIPTS

- Baiqiao Zhang, Junze Li, **Chunyang Luo**, Yuying Tang, Wenshuo Zhang, Shiyuan Piao, Dingdong Liu, Shuai Ma, and Xiaojuan Ma. *MemoSync: Co-Managing Developer-Agent Metamemory Across Contexts.* Submitted to **ACM CHI 2027**.

RESEARCH EXPERIENCE

OpenGRAIL: Open-World Task Grouping and Robot Allocation with Integrated LLM Planning Sept. 2025 – Present
Supervisor: Prof. Zhongkui Li

- Developed an LLM-based hierarchical planner that interprets natural-language instructions, environment context, and robot capabilities to generate dependency-aware task groups for large-scale, long-horizon tasks.
- Formulated heterogeneous robot allocation and scheduling within each group as a MILP under capability, temporal, spatial, and collaboration constraints.
- Integrated group-level solutions into globally executable plans, reducing planning time and improving planning accuracy at scale.

Manuscript in preparation for **ICRA 2027**.

Human-AI Collaboration for Long-Horizon Agent Systems Jul. 2026 – Sept. 2026
Summer Research, HKUST. Supervisor: Prof. Xiaojuan Ma

- Designed an interactive memory system that extracts and modularizes reusable memories from ongoing human-AI conversations.
- Developed structured mechanisms for storing and organizing memory modules by scope, type, lifecycle, and inter-memory relationships.
- Designed task-aware memory retrieval and scheduling to provide relevant context for downstream agent execution and preserve user intent across long-horizon interactions.

Multi-Robot Collaborative LTL Continuous Planning for Dynamic Environments Jul. 2025 – Present
Supervisor: Prof. Zhongkui Li

- Designed an online task decomposition and replanning framework for heterogeneous multi-robot LTL tasks under dynamic events.
- Built a heterogeneous multi-robot simulation platform (Unitree Go2, Scout) with task execution and state feedback interfaces.
- Benchmarked against static and periodic-replanning baselines on completion rate, planning time, and cumulative cost; currently extending to physical robots.

AWARDS & COMPETITIONS

- **Second Prize**, 22nd "Jiang Zehan Cup" Modeling and Computer Applications Competition, Peking University
- **Honorable Mention**, MCM/ICM 2026

SELECTED PROJECTS & TRAINING

RepoFlow: Repository-Level Workflow for AI Coding Agents Course Project, PKU
Designed persistent-context and structured agent-handoff mechanisms to improve task continuity in long-horizon coding agents.

Alibaba DAMO Academy – Embodied Intelligence Bootcamp Spring 2026
Built a robotic-arm visual-navigation demo integrating perception, control, and task execution.

TECHNICAL SKILLS

- **Programming & Tools:** Python, MATLAB, Linux, Git, LaTeX
- **Planning & AI:** LTL, PDDL, Classical Planning, Multi-Robot Task Allocation, LLM-based Planning & Reasoning, RL
- **Robotics:** ROS, Gazebo, RViz, AI2-THOR
- **Language:** CET-6 578, IELTS Overall 7.0